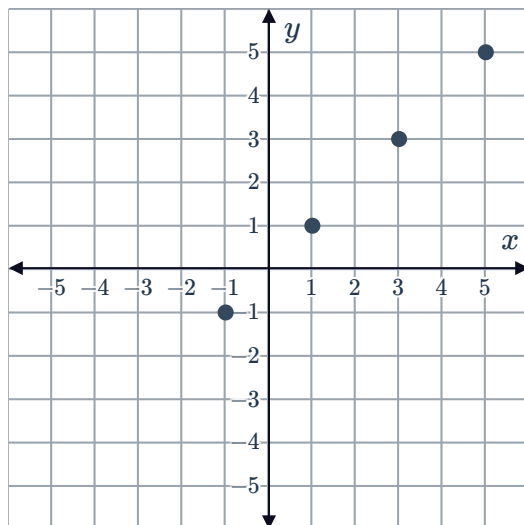


# Linear relationships

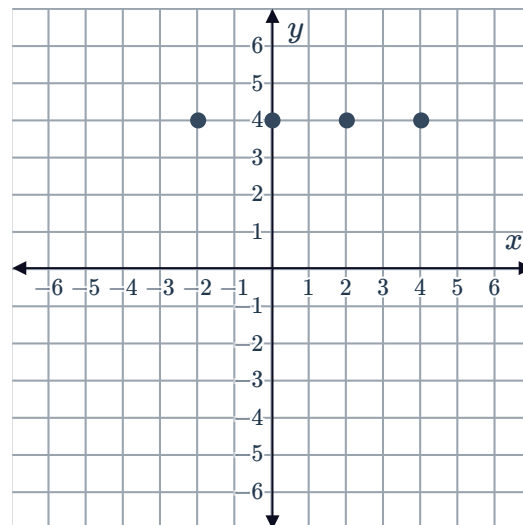


- 1 Do the points on each of the following number planes form a linear relationship? Explain your answer.

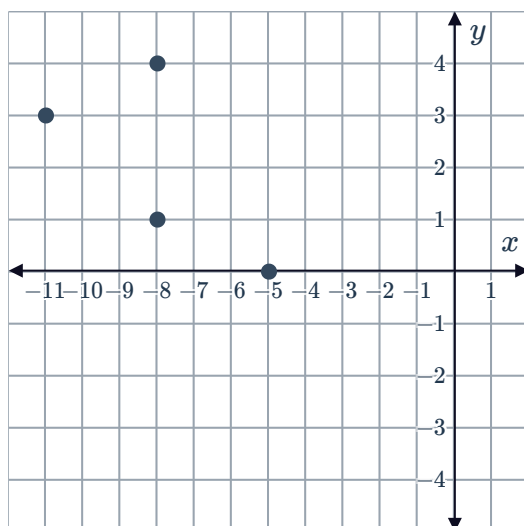
a



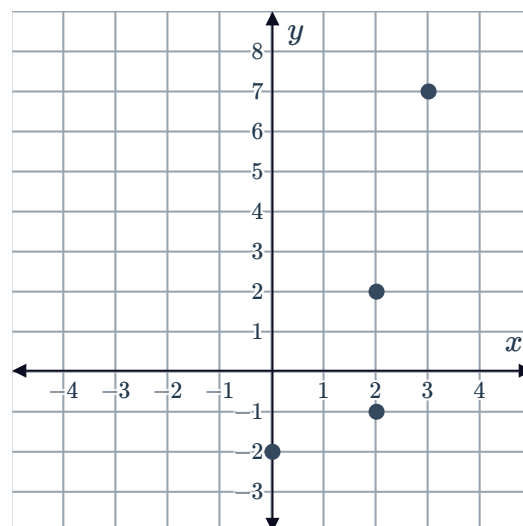
b



c



d



- 2 Consider the three points  $A(3, 6)$ ,  $B(1, 0)$ , and  $C(2, 3)$ .

a Plot the points on a number plane.

b Do the points on each of the following number planes form a linear relationship? Explain your answer.

- 3 Consider the three points  $X(3, -7)$ ,  $Y(4, -9)$ , and  $Z(1, -2)$ .

a Plot the points on the same number plane.

b Do the points on each of the following number planes form a linear relationship? Explain your answer.



## Equations, tables and graphs

4 For each of the following equations and their table of values:

- i Plot the points from the table on a number plane.
- ii Determine whether the points form a linear relationship.

a  $y = x + 5$

|     |   |   |   |   |
|-----|---|---|---|---|
| $x$ | 1 | 2 | 3 | 4 |
| $y$ | 6 | 7 | 8 | 9 |

b  $y = x$

|     |   |   |   |   |
|-----|---|---|---|---|
| $x$ | 1 | 2 | 3 | 4 |
| $y$ | 1 | 2 | 3 | 4 |

c  $y = 3$

|     |    |    |   |   |
|-----|----|----|---|---|
| $x$ | -2 | -1 | 0 | 1 |
| $y$ | 3  | 3  | 3 | 3 |

d  $y = x^2$

|     |    |    |   |   |
|-----|----|----|---|---|
| $x$ | -2 | -1 | 0 | 1 |
| $y$ | 4  | 1  | 0 | 1 |

e  $y = x + 4$

|     |    |    |    |    |
|-----|----|----|----|----|
| $x$ | -4 | -3 | -2 | -1 |
| $y$ | 0  | 1  | 2  | 3  |

f  $x = -6$

|     |    |    |    |    |
|-----|----|----|----|----|
| $x$ | -6 | -6 | -6 | -6 |
| $y$ | -6 | -3 | 0  | 3  |

g  $y = -x$

|     |    |    |    |   |    |
|-----|----|----|----|---|----|
| $x$ | -3 | -2 | -1 | 0 | 1  |
| $y$ | 3  | 2  | 1  | 0 | -1 |

h  $y = 4x$

|     |    |    |   |   |   |
|-----|----|----|---|---|---|
| $x$ | -2 | -1 | 0 | 1 | 2 |
| $y$ | -8 | -4 | 0 | 4 | 8 |

5 For each of the following equations:

- i Complete the table of values provided.
- ii Plot the points from the table on a number plane.

a  $y = 3x$

|     |    |   |   |   |
|-----|----|---|---|---|
| $x$ | -1 | 0 | 1 | 2 |
| $y$ |    |   |   |   |

b  $y = x - 5$

|     |   |   |   |   |
|-----|---|---|---|---|
| $x$ | 4 | 5 | 6 | 7 |
| $y$ |   |   |   |   |

c  $y = x + 3$

|     |   |   |   |    |
|-----|---|---|---|----|
| $x$ | 1 | 3 | 7 | 10 |
| $y$ |   |   |   |    |



d  $y = 2x + 7$

|     |    |    |   |   |
|-----|----|----|---|---|
| $x$ | -2 | -1 | 0 | 1 |
| $y$ |    |    |   |   |

e  $y = 3$

|     |    |    |    |   |
|-----|----|----|----|---|
| $x$ | -6 | -3 | -2 | 2 |
| $y$ |    |    |    |   |

f  $y = -2x + 1$

|     |    |    |   |   |
|-----|----|----|---|---|
| $x$ | -2 | -1 | 0 | 1 |
| $y$ |    |    |   |   |

6 Consider the given table of values:

a Plot the points in the table on a number plane.

|     |    |    |    |   |
|-----|----|----|----|---|
| $x$ | -3 | -2 | -1 | 0 |
| $y$ | 15 | 10 | 5  | 0 |

b Is the relationship between  $x$  and  $y$  linear?

c If we extend the table of values to include  $x = 1$ , find the corresponding value of  $y$ .

7 Consider the given table of values:

a Plot the points in the table on a number plane.

|     |    |    |   |   |
|-----|----|----|---|---|
| $x$ | -8 | -4 | 0 | 4 |
| $y$ | -2 | -1 | 0 | 1 |

b Is the relationship between  $x$  and  $y$  linear?

c If we extend the table of values to include  $x = 8$ , find the corresponding value of  $y$ .

8 Consider the given table of values:

a Plot the points in the table on a number plane.

|     |    |   |   |   |
|-----|----|---|---|---|
| $x$ | -1 | 0 | 1 | 2 |
| $y$ | -3 | 1 | 5 | 9 |

b Is the relationship between  $x$  and  $y$  linear?

c If we extend the table of values to include  $x = 3$ , find the corresponding value of  $y$ .

9 Consider the given table of values:

Does the ordered pair  $(-4, 8)$  satisfy the linear relationship between  $x$  and  $y$ ?

|     |    |    |    |    |
|-----|----|----|----|----|
| $x$ | -2 | -1 | 0  | 1  |
| $y$ | 2  | -1 | -4 | -7 |



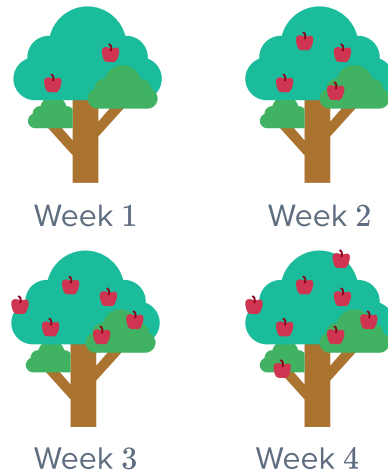
## Applications

**10** After each week, the number of apples on an apple tree increases according to the diagram:

**a** Complete the following table of values:

| Week       | 1 | 2 | 3 | 4 |
|------------|---|---|---|---|
| No. Apples |   |   |   |   |

- b** Plot the number of apples against each week on a number plane.
- c** Is the relationship between the number of weeks and the number of apples linear?

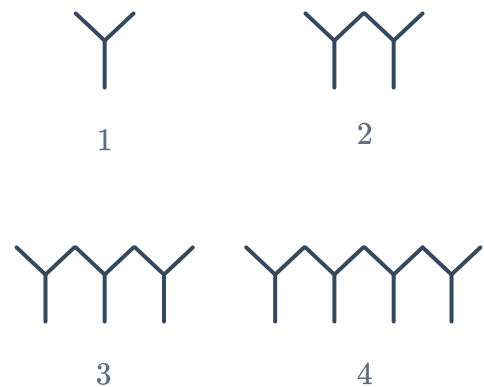


**11** The following pattern was created using matchsticks. Each figure is labelled with a number:

**a** Complete the following table of values:

| Figure number   | 1 | 2 | 3 | 4 |
|-----------------|---|---|---|---|
| No. Matchsticks |   |   |   |   |

- b** Plot the number of matchsticks used at each step on a number plane.
- c** Is the relationship between the step number and the number of matchsticks linear?

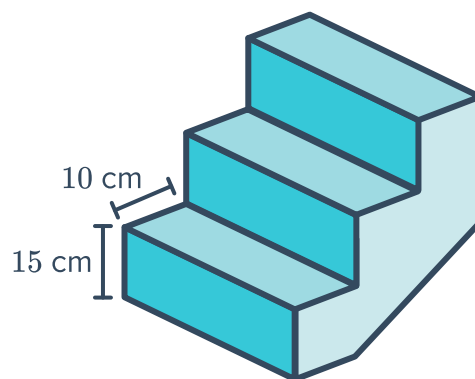


- 12** A staircase with 5 steps is shown below. The height of each step is 15 cm and the length of each step is 10 cm.



- a** Complete the following table of values:

| No. Steps                | 2  | 3 | 4 | 5 |
|--------------------------|----|---|---|---|
| Height of staircase (cm) | 30 |   |   |   |



- b** Plot the height of a staircase against the number of steps on a number plane.
- c** Is the relationship between the height of a staircase and the number of steps linear?
- d** Is it possible to have a staircase with height 105 cm with steps that have the same height and length as in the image?
- e** Complete the following table of values:

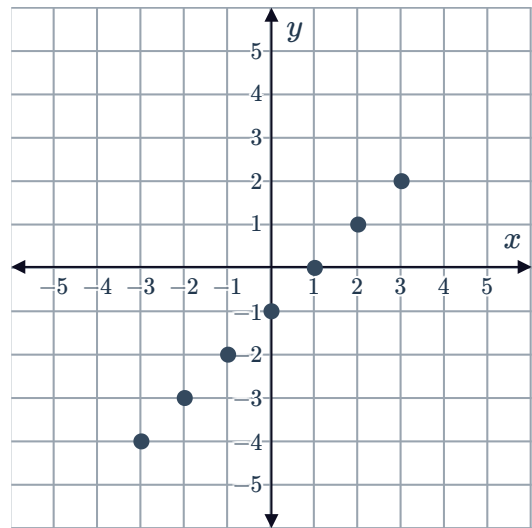
| No. Steps                | 2  | 3 | 4 | 5 |
|--------------------------|----|---|---|---|
| Length of staircase (cm) | 20 |   |   |   |

- f** Plot the length of a staircase against the number of steps on a number plane.
- g** Is the relationship between the length of a staircase and the number of steps in a staircase linear?
- h** Is it possible to have a staircase with length 70 cm with steps that have the same height and length as in the image?

- 13** A water tank initially starts with 65 L of water. Each day, 10 L of water is taken from the water tank.

- a** Plot the volume (in L) of water in the tank against the number days, from 0 to 4 days.
- b** Is the relationship between the volume of water in the water tank and the number of days linear?
- c** How many litres of water will there be after 5 days?

- 14 The points on the number plane show the relationship between water temperature,  $x$ , and surface air temperature,  $y$ :



- a Complete the table of values:

|                                                     |    |    |    |   |   |   |   |
|-----------------------------------------------------|----|----|----|---|---|---|---|
| Water Temperature in $^{\circ}\text{C}$ , $x$       | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| Surface Air Temperature in $^{\circ}\text{C}$ , $y$ |    |    |    |   |   |   |   |

- b Find the surface air temperature when the water temperature is  $4^{\circ}\text{C}$ .  
c Find the surface air temperature when the water temperature is  $14^{\circ}\text{C}$ .

- 15 Let the height of a candle be  $h$  cm. If the candle is lit, the height of the candle decreases according to the equation  $h = -2t + 9$ , where  $t$  is the time in minutes. A table of values is shown:

|                               |   |   |   |   |
|-------------------------------|---|---|---|---|
| Time<br>( $t$ mins)           | 0 | 1 | 2 | 3 |
| Height of candle<br>( $h$ cm) | 9 | 7 | 5 | 3 |

Plot the points in the table on a number plane.

- 16 A dead tree is 7 m tall and its height does not change over time. Let the height of the tree be  $y$  metres and the time passed be  $x$  years.

- a State whether the following coordinates satisfy the growth of the tree, where the  $x$ -coordinate is the time passed, and the  $y$ -coordinate is the height of the tree:

- i (3, 6)      ii (0, 7)      iii (7, 21)      iv (10, 1)  
v (4, 7)      vi (2, 7)      vii (7, 0)      viii (7, 7)

- b Plot the points from part (a) that satisfy the growth of the tree on a number plane.

- 17** Let the volume of water in a tank be  $V$  L,  the time taken to fill up the tank be  $t$  minutes. Suppose the tank is initially empty and the number of litres in the tank is always three times the number of minutes that have passed.

**a** Complete the table of values:

|                          |   |   |   |   |
|--------------------------|---|---|---|---|
| Time ( $t$ mins)         | 0 | 1 | 2 | 3 |
| Volume of water ( $V$ L) |   |   |   |   |

**b** Which of the following equations represents the relationship between the time taken,  $t$ , and the volume of the tank,  $V$ ?

**A**  $V = 3$     **B**  $V = 3t$     **C**  $V = 3t + 2$     **D**  $V = t + 3$

**c** Plot the points in the table on a number plane.

- 18** Let the height of a pot plant be  $y$  cm. As the pot plant grows, the height increases according to the equation  $y = 2t + 9$ , where  $t$  is the elapsed time in weeks.

**a** Complete the table of values:

|                               |   |   |   |   |
|-------------------------------|---|---|---|---|
| Time ( $t$ weeks)             | 0 | 1 | 2 | 3 |
| Height of pot plant ( $y$ cm) |   |   |   |   |

**b** Plot the points in the table on a number plane.

- 19** Max is 21 years old and Sebastian is 14 years old. Let  $M$  represent Max's age and  $S$  represent Sebastian's age.

**a** Write an equation for  $M$  in terms of  $S$ .

**b** State whether following ordered pairs, in the form  $(S, M)$ , satisfy the relationship between Max and Sebastian's ages:

- i** (14, 21)      **ii** (36, 29)      **iii** (20, 27)      **iv** (25, 25)  
**v** (24, 31)      **vi** (16, 23)      **vii** (38, 46)

**c** Plot the points from part (b) that satisfy the relationship between their ages on a number plane.